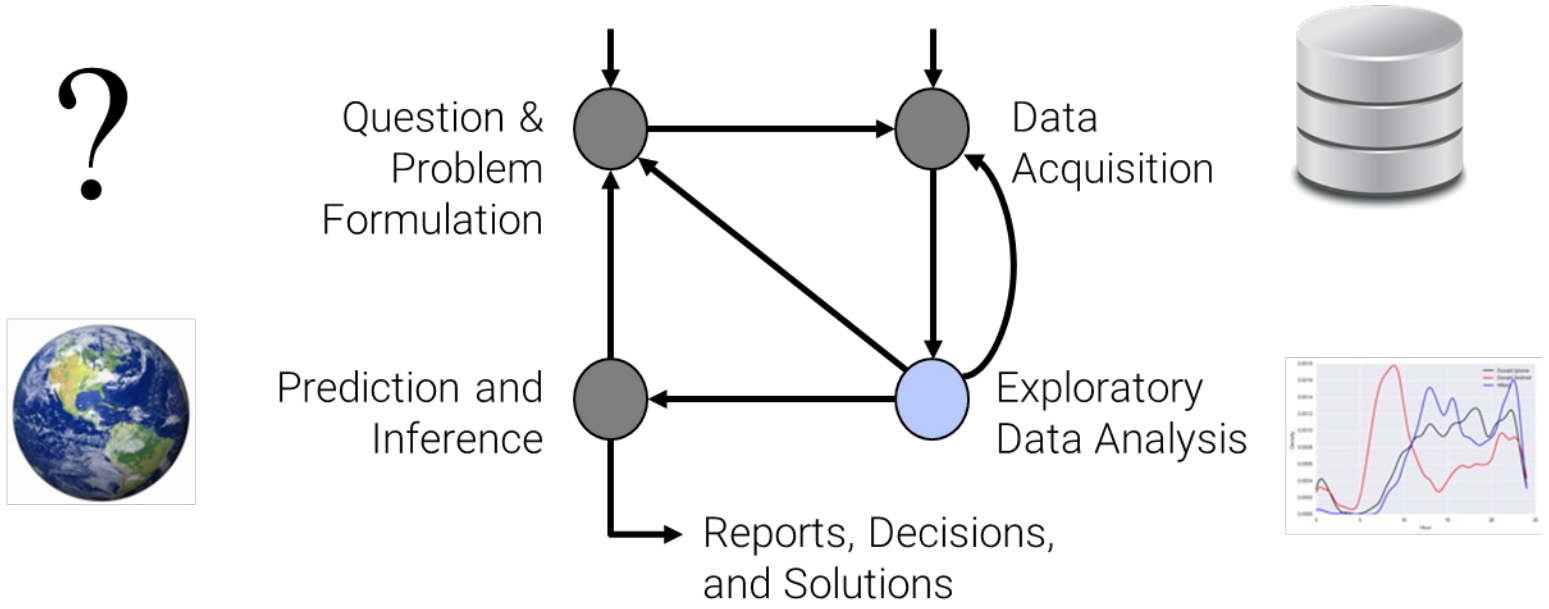
Introduction to Data Science

NumPy

Usman Nazir

Agenda

- What is NumPy?



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  - Linear algebra routines
  - Fourier transform and more.

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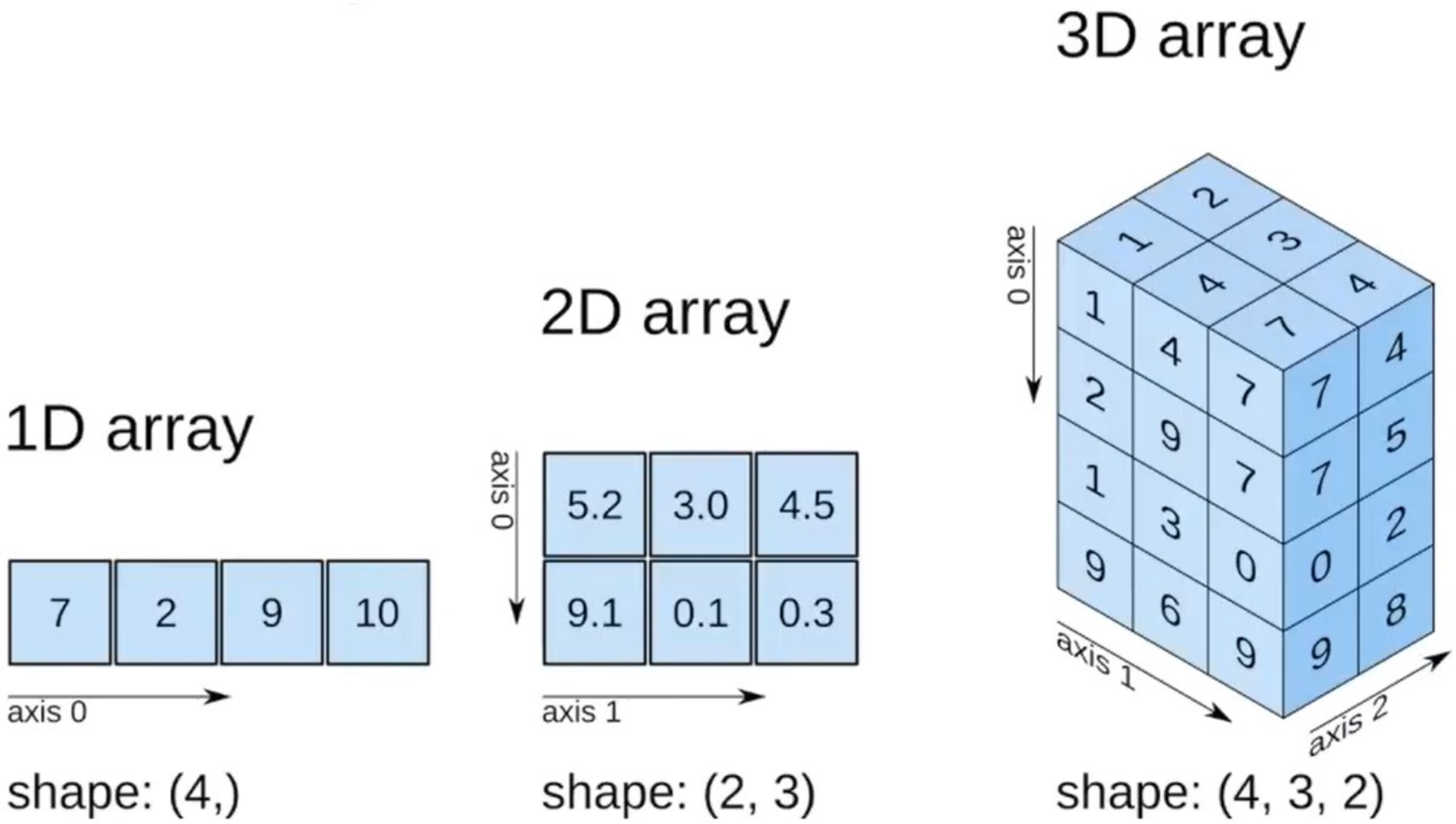
- Open source

N-dimensional arrays



Difference between Lists and NumPy arrays

Lists

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    - No type checking when iterating through objects
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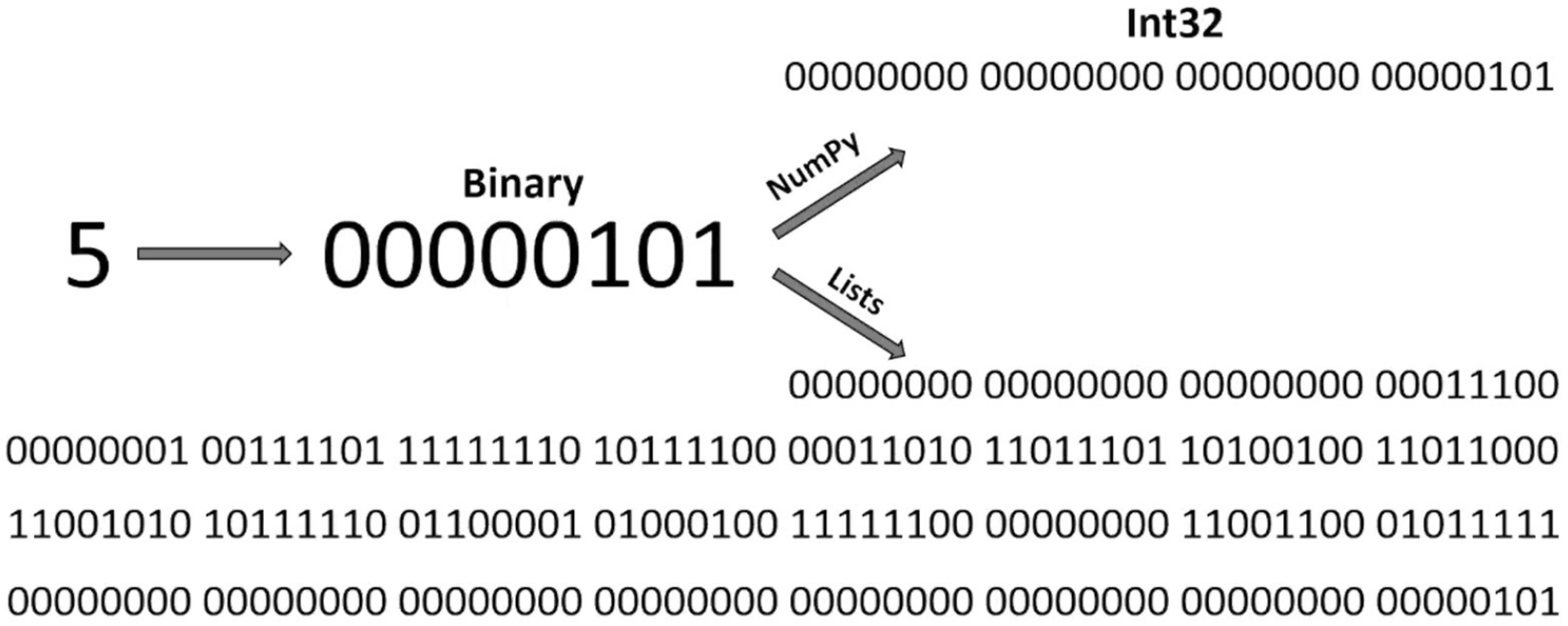

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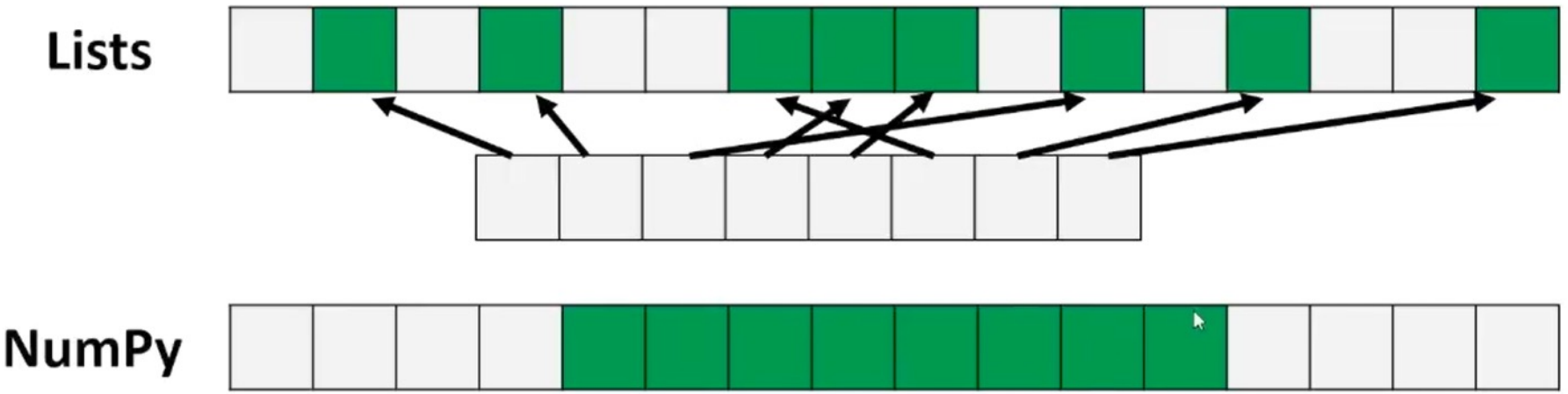

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  - SIMD (Single Instruction Multiple Data)
  - Effective Cache Utilization

How are Lists different from NumPy?

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  - A =np.array([1, 3, 5])
  - B = np.array([1, 2, 3])
  - A*B = np.array([1, 6, 15])

Applications of NumPy

- Mathematics (MATLAB replacement)
- Plotting (**Matplotlib**)
- Backend (**Pandas**, Connect 4, Digital Photography)
- Machine Learning applications' key library (Tensors)

Summary

- What is NumPy?